ITA03-MP ASSIGNMENT 1

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Course : Mobile Computing
Course Code : ITA0313

MOBILE COMPUTING
ASSIGNMENT 1 – CO1
CO1 Coverage: Cellular network concepts; Frequency reuse and interference in cells; Access channels and mobile network capacity; Fundamentals of mobile computing

Nature of Assignment:		Individual
Duration:	1	Week
Total Marks:		50

Submission Date: _____

Assignment Title: Design, Analyse and Optimise a Hexagonal Cellular Network for Spectrum Allocation, Interference Management and Scalable Coverage Planning in a University Township

(Total Marks – 50)

Duration:1week

Scenario — Cellular Coverage Planning for a University Township

An operator has been licensed to provide cellular service to a 12 km² university township containing academic blocks, hostels, residential colonies and a main arterial road. The operator holds a fixed allocation equivalent to 336 duplex voice channels (after control-channel overhead) and must plan a hexagonal-cell network. Subscriber demand will grow as the township expands.

You are the radio-planning team. You must investigate how spectrum can be shared across cells, how interference limits that sharing, and how the design must change as the subscriber base grows.

Planning data and constraints

- Subscribers: 1,000 → 10,000 → 20,000 → 100,000+ (planning horizon).
- Average busy-hour traffic per subscriber: 0.025 Erlang. Target grade of service: ≤ 2% call blocking.
- Minimum acceptable SIR: choose and justify a threshold between 12 dB (digital-style) and 18 dB (analogue-style).

- Path-loss exponent n: 2.5 (open campus) to 4 (dense built-up area); justify the values you adopt.
- Spectrum cannot be increased; only the network layout and radio parameters may change.
- Any planning effort (surveys, traffic estimation, number of sites) must be included in your cost analysis.

Design Challenge (Day 4):

A convocation / sports event will bring 25,000 attendees (0.05 Erlang each) into a 1 km² stadium zone for four hours. Re-evaluate your original design and decide whether to retain, modify or redesign it.

TASK 1 – Design Alternative Frequency-Planning Strategies 5 Marks

Guiding Enquiry

How can a fixed block of spectrum be shared across many cells so that every subscriber is served without intolerable co-channel interference — and in how many fundamentally different ways can this be done?

Investigate and interpret:

- Propose at least **three meaningfully different strategies** (for example fixed cluster reuse, sectorised reuse, dynamic/borrowed channel assignment, hierarchical macro–micro layering). The same strategy with different cluster sizes (N = 3 vs N = 7) does not count as two strategies.
- At least one strategy must be expressed as an **iterative procedure** (e.g., searching valid cluster sizes $N = i^2 + ij + j^2$ and assigning channel sets across the grid) and at least one as a **recursive/hierarchical procedure** (e.g., repeated cell splitting until demand is met).
- analyse the basic idea, the assumptions, and the way the **access channels** are shared (FDMA/TDMA/CDMA/OFDMA concept and control vs traffic channels).
- Provide pseudocode or a flowchart for each, name the design principle used (spatial reuse, trunking, spatial filtering), and justify why each deserves investigation.
- State the preprocessing required (traffic map, propagation survey, hex-grid layout) and who bears that cost.

Evidence Expected

Three distinct strategies with pseudocode/flowcharts, identified design principle, preprocessing requirements and justification for each.

TASK 2 – Identify the Governing Design Parameter 2 Marks

Guiding Enquiry

For each strategy, which single quantity most strongly controls capacity and interference, and why should the analysis be built around it?

Investigate and interpret:

- Identify the dominant parameter(s) for each strategy (e.g., cluster size N, reuse ratio Q = D/R, number of co-channel interferers, cell radius R, channels per cell).
- analyse where it appears in the strategy and why it is appropriate for efficiency analysis.
- interpret how this quantity must change as subscribers grow from 1,000 to 100,000+.

Evidence Expected

Correct identification and justification of the governing parameter for each strategy, linked to its behaviour.

TASK 3 – Derive the Capacity and Interference Functions Mathematically 6 Marks

Guiding Enquiry

Starting from hexagonal geometry, how do coverage, capacity and interference depend on the design choices you made?

Investigate and interpret:

- Derive the cluster-size relation, reuse distance $D = R\sqrt{(3N)}$ and reuse ratio Q, hexagonal cell area, and the number of cells needed to cover 12 km^2.
- Derive channels per cell (S/N), traffic per cell at 2% blocking (Erlang-B reasoning), and total system capacity as a function of N and R.
- Derive the co-channel SIR expression for each strategy (first-tier interferers; effect of sectoring on the number of interferers).
- Combine **planning cost** (survey, number of sites) with **operating capacity** to obtain an overall cost–capacity relation for each strategy.

- Show every step; stating a final formula without derivation will not earn full credit.

Evidence Expected

Clear derivations of capacity, number of cells and SIR for each strategy, including planning cost and overall cost–capacity relation.

TASK 4 – Analyse Best, Average and Worst-Case Behaviour 4 Marks

Guiding Enquiry

Where in the cell, and at what time of day, does each strategy perform best, typically, and worst — and why?

Investigate and interpret:

- Identify the position of the mobile (near the base station, mid-cell, cell corner) that yields best, average and worst SIR, and the geometry of interferers in each case.
- Identify the traffic condition (off-peak, average, busy-hour, event surge) that yields each blocking behaviour.
- analyse how shadowing and a changing path-loss exponent shift each case.
- Support each case with calculations, not just a table of values.

Evidence Expected

Justified best/average/worst analysis for SIR and blocking, supported by mathematical reasoning.

TASK 5 – Scaling and Sensitivity Analysis 3 Marks

Guiding Enquiry

How do capacity, interference and the number of cells scale as demand and the environment change, and at what point does each strategy stop being practical?

Investigate and interpret:

- Express proportionalities (e.g., capacity $\propto 1/N$, SIR $\propto N^{(n/2)}$, cells $\propto 1/R^2$) and compare growth of the strategies.
- Interpret what happens when subscribers grow from 1,000 to 100,000+ and when n varies from 2.5 to 4.

- Identify a break-even scale beyond which a different strategy becomes preferable.

Evidence Expected

Correct scaling relations and an interpretation tied to the township application.

TASK 6 – Formulate the Recurrence for Repeated Cell Splitting 3 Marks

Guiding Enquiry

If overloaded cells are split repeatedly, how can the number of cells, capacity, transmit power and handoff load after k rounds be expressed in terms of round $k - 1$?

Investigate and interpret:

- State the original problem size (cells and demand) and the subproblem produced by each split (radius $R \rightarrow R/2$).
- Identify the number of new cells per split, the work/cost per round, and the base case (the initial layout).
- Formulate recurrences for number of cells, offered capacity and required transmit power ($P' = P/2^n$), derived from your own splitting procedure.
- Allow for selective splitting (only a fraction of cells is overloaded) as well as uniform splitting.

Evidence Expected

Correct recurrence(s) with clearly identified components and base case.

TASK 7 – Solve the Recurrence and Interpret the Result 4 Marks

Guiding Enquiry

How many splitting rounds are needed to carry the forecast demand, and what do the closed-form results say about cost and mobility?

Investigate and interpret:

- Solve using iteration/substitution (or another suitable method) and justify the method.
- Derive the minimum number of rounds $k = \lceil \log_4(D/C_0) \rceil$ for uniform splitting and its selective counterpart.

- Interpret the result: growth in sites, reduction in transmit power, and the increase in handoff rate as radius shrinks.

Evidence Expected

Complete solution with method justification and an interpretation of the closed form.

TASK 8 – Implement and Test the Planning Strategies 4 Marks

Guiding Enquiry

Can a simulator built from your own strategies reproduce the behaviour predicted by your mathematics?

Investigate and interpret:

- Implement all strategies (e.g., Python/NumPy): hex-grid generator, channel assignment, Monte-Carlo SIR evaluator and Erlang-B calculator.
- Use scaled datasets: 1,000 → 10,000 → 20,000 → 100,000 subscribers with the same random seeds across strategies.
- Include test cases: N = 3, 4, 7, 12; n = 2.5, 3, 4; sectored and unsectored layouts.
- Sanity-check with a known case (N = 7, n = 4 under the six-equidistant-interferer approximation gives about 18.7 dB).

Evidence Expected

Working code, reproducible procedure and correctness checks aligned to the analysed strategies.

TASK 9 – Record and Present Experimental Performance3 Marks

Guiding Enquiry

Which measurements best expose the differences between your strategies, and how should they be presented?

Investigate and interpret:

- Record SIR distribution (mean, 5th percentile), outage probability, blocking probability, channels per cell, sites required and run-time.
- Present tables and plots such as SIR CDFs, blocking vs offered load and capacity vs N.
- Ensure the presentation allows direct comparison between strategies.

Evidence Expected

Systematically collected data presented in comparable tables and graphs.

TASK 10 – Compare Theory with Simulation 4 Marks

Guiding Enquiry

Where do the simulated results agree with your analytical predictions, where do they differ, and what analyses the gap?

Investigate and interpret:

- Compare analytical and simulated SIR and blocking for each strategy.
- Identify sources of deviation: equidistant-interferer approximation, shadowing, finite tiers, mobile-position distribution, Poisson-arrival and infinite-source assumptions in Erlang-B, integer rounding of N and channels.
- Decide which strategy scales best and defend this using both results.

Evidence Expected

A reasoned comparison of theory and simulation with analyseed discrepancies.

TASK 11 – Recommend and Defend the Final Design 4 Marks

Guiding Enquiry

Which design would you actually deploy in the township, and what evidence makes it better than the alternatives?

Investigate and interpret:

- State the recommended strategy, parameters (N, R, sectoring) and expected capacity/SIR/blocking.
- analyse why it is superior to the alternatives using your analysis and simulation.
- interpret planning cost, trade-offs (trunking efficiency vs interference, handoff load vs capacity) and limitations.
- Judge whether the design remains practical as the township grows past 100,000 subscribers.

Evidence Expected

A technically justified recommendation supported by analysis and simulation; choosing the highest-capacity option alone is not sufficient.

DAY-4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design 8 Marks

Changed Requirement

A convocation / sports event will bring 25,000 attendees, each generating 0.05 Erlang, into a 1 km² stadium zone for four hours. Outside the event, demand stays as originally forecast. Event days occur roughly twice a month. Re-evaluate your original design under the changed conditions.

You may:

RETAIN → MODIFY → or REDESIGN

your original solution.

A. Impact Analysis – 2 Marks

Guiding Enquiry

What exactly has changed, and which of your original assumptions no longer hold?

- Estimate the new offered traffic density and compare it with the design density.
- Identify affected assumptions (uniform traffic, steady demand, permanent capacity).
- analyse why the short duration and predictability of the surge matter for the choice of solution.

Evidence Expected

A clear analysis of how the surge affects the original design.

B. Evaluation of the Original Design – 2 Marks

Guiding Enquiry

How would your recommended design behave on event day?

- Estimate blocking probability and interference problems in the stadium zone using your own model.
- Assess whether the design can be reused unchanged and what fails first.
- Identify whether an alternative strategy handles the surge more economically.

Evidence Expected

A technically supported evaluation of the original design under the surge.

C. Adaptation or Redesign – 2 Marks

Guiding Enquiry

Should the township network be altered permanently, temporarily or not at all?

- Choose retain, modify or redesign (e.g., micro-cells/cell splitting in the stadium zone, 60° sectoring, temporary cells-on-wheels, dynamic channel borrowing).
- Describe the revised strategy and what changes in reuse pattern, power and handoffs.
- Determine the effect on capacity, SIR and planning cost.

Evidence Expected

A design response with clear radio-planning reasoning and quantified effect.

D. Evidence-Based Defence – 2 Marks

Guiding Enquiry

Why is your final decision the right one, given cost, frequency of events and the evidence you generated?

- Use analytical results, simulation results, surge size and duration, planning cost and trade-offs.
- Show that the decision follows logically from your evidence.

Evidence Expected

A well-supported final decision integrating theory, simulation and the changed constraints.

MARK DISTRIBUTION

Task	**Marks**
Task 1 – Design Alternative Frequency-Planning Strategies	5
Task 2 – Identify the Governing Design Parameter	2
Task 3 – Derive the Capacity and Interference Functions Mathematically	6
Task 4 – Analyse Best, Average and Worst-Case Behaviour	4
Task 5 – Scaling and Sensitivity Analysis	3
Task 6 – Formulate the Recurrence for Repeated Cell Splitting	3
Task 7 – Solve the Recurrence and Interpret the Result	4

Task 8 – Implement and Test the Planning Strategies	4
Task 9 – Record and Present Experimental Performance	3
Task 10 – Compare Theory with Simulation	4
Task 11 – Recommend and Defend the Final Design	4
Subtotal	**42**
Task 12 – Day-4 Design Challenge	8
TOTAL	**50**

DESIGN BASIS, ASSUMPTIONS AND METHOD

This report treats the supplied assignment brief as the governing specification. The township area is 12 km², the licensed allocation is 336 duplex voice traffic channels after control-channel overhead, busy-hour demand is 0.025 Erlang per subscriber, and the target blocking probability is at most 2%. Subscriber cases are 1,000, 10,000, 20,000 and 100,000+. The design challenge adds 25,000 event attendees at 0.05 Erlang each in a 1 km² stadium zone for four hours, with events about twice per month.

For a transparent base design, select a nominal hexagonal cell radius R = 0.50 km and cluster size N = 7. The radius is a planning assumption rather than a measured coverage result. It gives manageable site density while leaving room for local densification. Use path-loss exponent n = 3.5 as the township-wide planning value: it lies between the open-campus value 2.5 and dense-built value 4.0. For open academic grounds use n≈2.5–3; for buildings, hostels and street canyons test n=4. Use a minimum planning SIR of 18 dB for a conservative voice-quality target, while also reporting the 12 dB digital-style reference. Real deployment must validate these assumptions through drive tests and propagation surveys.

A critical distinction is maintained throughout: the Erlang-B results below are analytical capacity estimates, not measurements from a live network. Likewise, the simulator supplied in Task 8 is a reproducible baseline model; its output must be generated by running the code in the student's environment. No unrun simulation values are presented as measured results. Site counts are rounded upward, and cell-edge overlap, terrain, indoor penetration and control-channel details are treated as planning margins rather than silently assumed away.

The fixed allocation is shared spatially. For cluster size N, each cell receives approximately S/N channels, where S=336. Because channels are indivisible, use floor(336/N) for a conservative equal allocation and distribute any remainder among cells only if the reuse plan preserves separation. For N=7, there are 48 channels per cell. The remaining six channels are not treated as additional traffic capacity in this conservative equal-set model; an operator could allocate them through a carefully engineered nonuniform plan.

TASK 1 – STRATEGY 1: FIXED CLUSTER FREQUENCY REUSE

Fixed cluster reuse divides the 336 traffic channels into N disjoint channel groups. A cluster contains N neighbouring hexagonal cells, and each group is assigned once per cluster. The pattern repeats over the township. This is spatial reuse: the same frequency set is reused only after a separation distance intended to keep co-channel interference below the design limit. The baseline uses N=7, giving 48 channels per cell. Alternative valid hexagonal cluster sizes include N=3, 4, 7, 9, 12, 13 and others of the form i^2+ij+j^2; N=3, 4, 7 and 12 are explicitly useful test cases.

The geometric assumptions are ideal hexagons, uniform reuse, equal-power base stations, a regular grid, and first-tier co-channel interferers. Real cells are irregular, so this is a first-order planning model. Access may be implemented with FDMA/TDMA voice channels or as logical resource blocks in a modern OFDMA system. Control channels carry broadcast, paging, access and signalling functions; traffic channels carry user voice/data. The given 336 channels are already net of control overhead, so control resources must not be subtracted a second time.

Iterative planning procedure: enumerate valid N values, calculate channels per cell and reuse distance, estimate SIR, reject candidates below the chosen threshold, calculate Erlang-B capacity, then select the lowest-cost candidate meeting demand. Pseudocode: SET S=336; FOR each valid N in {3,4,7,12}: c=floor(S/N); Q=sqrt(3N); D=Q*R; estimate SIR from n and interferer geometry; Amax=ErlangB_inverse(c,0.02); calculate cells=ceil(12/Ahex); calculate supported subscribers; IF SIR≥18 dB and capacity≥forecast demand THEN mark feasible; END; compare feasible candidates by cost and handoff burden.

Preprocessing includes a population/traffic heat map, building-height and clutter map, candidate-site survey, hexagonal grid generation, frequency-neighbour list, and propagation measurements. The operator bears radio-planning and site-survey costs; the university or township authority may share access and wayleave costs under a commercial agreement. This strategy deserves investigation because it is simple, predictable, auditable and inexpensive to operate. Its weakness is rigid allocation: a quiet cell cannot automatically lend unused channels to a crowded neighbour.

TASK 1 – STRATEGY 2: SECTORISED CELLS

Sectorisation retains a macrocell site but divides its coverage into directional sectors using separate antennas and radio chains. A common arrangement is three 120° sectors; a more aggressive arrangement uses six 60° sectors. Directional antennas reduce the number and strength of dominant co-channel interferers seen by a sector. This is spatial filtering, not an increase in licensed spectrum. The same channel group can be reused at a shorter distance only when antenna orientation, downtilt, sidelobes and neighbour planning support the interference target.

For the first-order ideal model, an omnidirectional cell has six dominant first-tier co-channel interferers. A 120° sector often reduces the dominant set to approximately two, while a 60° sector can reduce it further; actual counts depend on site geometry and antenna patterns. Do not assume that sectoring automatically triples or sextuples total spectrum. It partitions the cell into directional coverage areas and reduces interference; sector capacity is constrained by the same aggregate licensed channel pool and by sector-specific traffic imbalance.

Pseudocode: FOR each candidate site: divide azimuth into 3 or 6 sectors; assign antenna pattern and azimuth; build sector neighbour lists; allocate channel/resource groups consistent with reuse constraints; simulate users at random locations; compute desired and interfering received powers including antenna gain; estimate outage and blocking per sector; adjust azimuth, downtilt and reuse plan until all sectors satisfy SIR and traffic limits. The procedure is iterative because antenna parameters and frequency assignments are tuned together.

Assumptions include known antenna patterns, adequate isolation between sectors, calibrated power, and manageable traffic balance. Sectorisation is attractive around the main arterial road and dense academic/hostel clusters where users arrive in directional corridors. Preprocessing requires traffic-direction observations, azimuth/site visibility checks, antenna-pattern files and interference measurements. The operator bears antenna, radio-unit, installation and optimisation costs. Sectoring is worth investigating because it can improve SIR without obtaining more spectrum, but it increases equipment, configuration complexity and handoff boundaries.

TASK 1 – STRATEGY 3: DYNAMIC CHANNEL ASSIGNMENT AND BORROWING

Dynamic channel assignment (DCA) selects a channel from a shared pool when a call arrives, subject to current interference and occupancy constraints. Borrowing is a related method: a congested cell temporarily uses an idle channel from a neighbouring cell, provided the donor and its co-channel neighbours remain protected. This uses trunking more flexibly than fixed allocation. The design principle is statistical multiplexing: capacity is placed where demand occurs rather than permanently reserving the same number of channels in every cell.

The central assumption is that a central controller or distributed scheduler has timely information about channel occupancy, interference, neighbour relationships and call demand. A candidate channel is accepted only if it does not violate the reuse/interference rule. In an OFDMA network the analogous operation is dynamic scheduling of time-frequency resource blocks; in legacy FDMA/TDMA it is channel selection. Control channels remain available for access and coordination, while the 336 net traffic channels are dynamically assigned.

Pseudocode: WHEN a call request arrives in cell x: construct candidate set from free local channels plus borrowable neighbour channels; FOR each candidate channel f: check co-channel separation and measured interference; calculate predicted SIR; discard f if predicted SIR<18 dB; IF candidates remain choose the one with lowest interference / lowest borrowing cost; reserve f and update neighbour occupancy; ELSE block or queue according to service policy. WHEN call ends: release f and return borrowed channel to donor pool. Periodically update interference and load estimates.

DCA requires a live traffic map, a channel occupancy database, neighbour and interference measurements, and controller/backhaul support. The operator pays for planning, software, signalling and monitoring. It deserves investigation because busy-hour demand is spatially uneven and the event stadium produces a short, concentrated surge. Its weaknesses are control complexity, coordination delay and the risk of unstable borrowing if adjacent cells simultaneously seek the same resource. A practical design uses fixed reuse as a safe baseline and permits borrowing only under explicit interference constraints.

TASK 1 – STRATEGY COMPARISON AND PREPROCESSING

The three strategies are genuinely different. Fixed reuse makes a static spatial partition of channels; sectorisation changes the antenna pattern and the interference geometry; dynamic assignment changes the time-varying allocation decision. A fourth useful layer is hierarchical macro–micro deployment: macrocells provide broad coverage and micro/pico cells add local capacity in high-demand zones. This is a recursive spatial densification approach and is developed in Tasks 6 and 7.

Comparison: fixed reuse is easiest to plan and has predictable SIR, but wastes capacity when traffic is uneven. Sectorisation improves directional isolation and can improve cell-edge SIR, but costs more per site and may create sector imbalance. Dynamic assignment improves trunking efficiency and handles local peaks, but needs accurate measurements and robust coordination. A hybrid is normally stronger than any one method alone: fixed macro reuse for coverage, sectorisation in high-interference corridors, and constrained dynamic borrowing for temporary imbalance.

Preprocessing sequence: (1) collect subscriber density by building and time of day; (2) estimate busy-hour Erlangs using subscribers×0.025; (3) map buildings, terrain, road corridors and indoor-heavy zones; (4) select candidate site coordinates and heights; (5) construct a hex-grid approximation; (6) conduct propagation and interference surveys; (7) generate channel/sector neighbour lists; (8) validate through a pilot and drive testing. Planning must include staff time, survey equipment, access permissions, software, site acquisition and optimisation.

The operator is primarily responsible for the radio network and its engineering cost. The university can reduce cost by providing rooftops, utility access and site permissions. The township authority may coordinate permits and road access. These responsibilities should be documented because site acquisition delay can dominate the deployment schedule even when the radio design is mathematically feasible.

TASK 2 – GOVERNING DESIGN PARAMETERS

For fixed reuse, the governing parameter is cluster size N, equivalently reuse ratio $Q=D/R=\sqrt{(3N)}$. Increasing N increases co-channel separation and usually improves SIR, but reduces the number of channels available per cell, $c \approx S/N$. Thus N controls the principal capacity–interference trade-off. It appears directly in the cluster geometry, channel partition and SIR expression.

For sectorisation, the dominant parameter is the effective number and strength of co-channel interferers, represented by the sector antenna pattern and sector angle. Narrower sectors suppress more off-axis energy, improving SIR, but the benefit depends on sidelobe level, azimuth alignment and traffic distribution. The number of sectors alone is not a reliable capacity multiplier.

For dynamic assignment, the governing parameter is the number of simultaneously available interference-safe channels in the local candidate pool. This depends on current occupancy, neighbour constraints and the measured SIR margin. The parameter varies with time, unlike fixed N. For hierarchical splitting, the cell radius R (and resulting site density $\propto 1/R^2$) governs local capacity and handoff rate.

At 1,000 subscribers, total offered traffic is 25 Erlangs and the baseline network has ample aggregate capacity. At 10,000 it is 250 Erlangs; at 20,000 it is 500 Erlangs; at 100,000 it is 2,500 Erlangs. Fixed reuse may retain N=7 initially, but at high load the network needs more spatial reuse through smaller cells, sectorisation or controlled DCA. N should not simply be reduced without checking SIR. For dynamic assignment, the safe pool becomes increasingly occupied, so the controller needs more local resources or additional microcells. The governing parameter is useful because it exposes exactly where each design trades capacity for interference protection.

TASK 3 – HEXAGONAL GEOMETRY AND CLUSTER SIZE DERIVATION

A regular hexagonal cell with centre-to-vertex radius R has area equal to six equilateral triangles of side R. One triangle has area $(\sqrt{3}/4)R^2$, so $A_{hex}=6(\sqrt{3}/4)R^2=(3\sqrt{3}/2)R^2$. With R=0.50 km, $A_{hex}=(3\sqrt{3}/2)(0.25)=0.6495$ km². The ideal number of cells to cover 12 km² is 12/0.6495=18.48, hence at least 19 cells. Allowing overlap, boundaries and indoor coverage, an operational design may require additional sites.

Hexagonal reuse clusters are generated by moving i cells along one hexagonal axis and then turning 60° and moving j cells. The squared centre separation in units of cell spacing is i^2+ij+j^2. Therefore $N=i^2+ij+j^2$. Examples: (i,j)=(1,1) gives N=3; (2,0) gives N=4; (2,1) gives N=7; (2,2) gives N=12. Not every positive integer is a valid ideal hexagonal cluster size.

The co-channel reuse distance is $D=R\sqrt{(3N)}$. Dividing by R gives reuse ratio $Q=D/R=\sqrt{(3N)}$. For N=7, $Q=\sqrt{21}=4.583$ and $D=0.50\times4.583=2.291$ km. For N=3, Q=3.000 and D=1.500 km. For N=4, Q=3.464 and D=1.732 km. For N=12, Q=6.000 and D=3.000 km. Larger N increases separation but allocates fewer channels to each cell.

For a finite township, the ideal 19-cell count is a geometric lower bound. A real hexagonal grid will have partial edge cells and overlapping coverage. Site count must be rounded up and then adjusted using link budgets, terrain, building penetration and road coverage. If a smaller R is chosen, the number of cells rises approximately as $1/R^2$. This is the basic mechanism for capacity densification when the spectrum allocation cannot grow.

TASK 3 – CHANNELS PER CELL AND ERLANG-B CAPACITY

The net traffic channel pool is S=336. Under equal fixed reuse, the conservative integer channel count is c=floor(S/N). For N=3, c=112; N=4, c=84; N=7, c=48; N=12, c=28. The Erlang-B recursion for c circuits and offered traffic A is B(0,A)=1 and B(c,A)=A·B(c−1,A)/(c+A·B(c−1,A)). It models call loss when all circuits are busy, assuming Poisson arrivals, exponential holding times and no waiting queue.

To find the maximum offered traffic at 2% blocking, invert Erlang-B numerically. The calculated approximate values are: 112 channels → 99.62 Erlangs; 84 channels → 72.53 Erlangs; 48 channels → 38.39 Erlangs; 28 channels → 20.15 Erlangs. These are per-cell offered loads, not carried traffic. At the 2% target, carried traffic is approximately A(1−B), so for 48 channels it is 38.39×0.98=37.62 Erlangs.

For the N=7 baseline, 19 cells provide a theoretical aggregate offered capacity of 19×38.39=729.4 Erlangs at 2% blocking, if every cell is equally loaded and channels are independently pooled within cells. The subscriber capacity is A_total/0.025=29,176 subscribers. This is an optimistic aggregate bound because real traffic is nonuniform and the edge cells may not be fully loaded. At 1,000, 10,000 and 20,000 subscribers the mean offered load per cell is 1.32, 13.16 and 26.32 Erlangs respectively; at 100,000 it is 131.58 Erlangs per cell, well above the 38.39-Erlang per-cell target.

For the baseline geometry, N=7 and R=0.5 km, the number of cells is approximately ceil(12/Ahex)=19. Total system capacity can be written Csys(N,R)=M(R)·A_B(floor(S/N),0.02), where M(R)=ceil(12/Ahex(R)) and A_B is the inverse Erlang-B offered load. This formula reveals the interaction: decreasing R increases M; increasing N improves reuse separation but reduces c and per-cell traffic capacity.

TASK 3 – SIR DERIVATION AND SECTORING EFFECT

Received power under a distance power law is proportional to $P \cdot d^{(-n)}$, where P is transmit power and d is distance. For a mobile at distance r from its serving base station and six first-tier co-channel interferers at approximately distance D, the equal-power approximation gives $SIR=(r^{(-n)})/(6D^{(-n)})=(1/6)(D/r)^n$. At the cell edge, take $r \approx R$ and $D/R=\sqrt{(3N)}$; hence $SIR \approx (1/6)(\sqrt{(3N)})^n=(3N)^{(n/2)}/6$.

For N=7 and n=4, $SIR=(21^2)/6=73.5$ in linear units. In dB, $10\log10(73.5)=18.66$ dB, matching the assignment's stated sanity-check value of about 18.7 dB. This is an ideal first-tier estimate, not a guarantee for every mobile. At n=3.5, $SIR=(21^{1.75})/6 \approx 48.8$, or about 16.9 dB, which is below the selected 18 dB threshold. Dense clutter can therefore require sectoring, careful power control, a larger effective reuse distance or a different antenna/site layout.

With m equal dominant interferers, the approximation becomes $SIR \approx (D/R)^n/m=(3N)^{(n/2)}/m$. If an ideal 120° sector reduces the dominant set from six to two, the gain is $10\log10(6/2)=4.77$ dB. If a 60° sector reduced it to one dominant interferer, the ideal gain relative to six would be 7.78 dB. These gains are upper-bound illustrations: antenna sidelobes, reflections and non-equidistant interferers reduce them.

For DCA, SIR is evaluated channel-by-channel: $SIR=P_{serv} \cdot g_{serv}$ divided by the sum of received co-channel powers from cells currently using that resource. A channel is admitted only when predicted SIR meets the threshold. For sectorised DCA, antenna gain $G(\theta)$ multiplies each received signal term. Thus fixed reuse has a geometry-based closed form, while sectoring and DCA require an antenna/occupancy-aware interference sum.

TASK 3 – COST–CAPACITY RELATION

A planning cost model should distinguish one-time and recurring costs. Let Csurvey be traffic mapping, propagation survey and drive-test cost; Csite be acquisition and installation cost per site; Cradio be radio/antenna equipment cost per site; Copt be software and optimisation cost; and Osite be annual operation/maintenance cost per site. For M cells, a simple deployment cost is Cfixed=Csurvey+M(Csite+Cradio)+Copt. Annual operating cost is O=M·Osite+Obackhaul+Oenergy. The values must be obtained from local quotations; the assignment provides no rupee rates, so inventing prices would be misleading.

For fixed reuse, Csys(N,R)=M(R)A_B(floor(S/N),0.02). A useful efficiency metric is capacity per capital cost: ηfixed=Csys/Cfixed. Increasing N may increase SIR but reduce per-cell channels; decreasing R increases M and therefore both capacity and site cost. For sectorisation, add Csector per sector/site and additional maintenance: Csect=Cfixed+M·(Csector-unit cost×number of sectors). Its capacity benefit is indirect through reduced interference and possibly a feasible smaller N.

For dynamic assignment, add controller, measurement and software cost Cctrl. Its effective capacity is not simply a larger S; it improves the utilisation of the same pool by reducing idle-channel fragmentation and borrowing unused local capacity. A conceptual expression is Cdyn≈M·A_B(c_eff,0.02), where c_eff is the interference-safe channel availability after occupancy and borrowing constraints. c_eff must be estimated from a time-varying simulation, not assumed equal to all 336 channels in every cell.

A fair economic comparison uses lifecycle cost over T years: LCC=Cinitial+Σ(year=1 to T) Oyear/(1+rdisc)^year. Compare cost per carried Erlang or cost per supported subscriber at the same blocking and SIR targets. The cheapest option is not necessarily the one with the lowest site count: if interference causes outage, additional coverage sites or repeated optimisation may be required. A pilot survey reduces this uncertainty and should be budgeted before full deployment.

TASK 4 – BEST, AVERAGE AND WORST SIR

Best SIR normally occurs close to the serving base station, because desired received power rises as $r^{(-n)}$, while interferers remain comparatively distant. In the ideal six-interferer model, at $r=0.5R$ the SIR is multiplied by $(R/0.5R)^n=2^n$ compared with the cell edge. For $n=3.5$ this is $2^{3.5}=11.31$, a gain of 10.54 dB. Starting from the ideal edge estimate 16.9 dB, the near-site estimate is about 27.4 dB.

An average mobile position may be represented by $r\approx0.65R$ as a simple illustrative test point, not an exact spatial average. The SIR gain over the edge is $(1/0.65)^{3.5}\approx4.48$, or 6.51 dB. Thus the illustrative SIR is about 23.4 dB. A rigorous average requires integrating over the user distribution and actual interferer locations. The cell corner or edge is usually worst because the desired base station is farther away and one or more neighbouring co-channel cells may be closer than the six-equidistant approximation assumes.

At the cell edge with $N=7$, $n=3.5$, six equal interferers, $SIR\approx16.9$ dB. With 120° sectoring and two dominant interferers, the ideal estimate rises by 4.77 dB to about 21.7 dB. This clears the 18 dB target in the simplified model. But an antenna's rear and side lobes, reflections and shadowing can lower the actual value. Therefore sectoring must be verified with measured patterns and drive tests.

Shadowing adds a random local variation to received power. A mobile behind a building may experience lower desired power, while an interferer with a clear path may dominate. Increasing n from 2.5 to 4 generally improves distance-based reuse isolation for a fixed geometry, but dense built-up areas also introduce blockage and shadowing variance. Consequently, the path-loss exponent alone cannot predict all worst cases.

TASK 4 – TRAFFIC AND BLOCKING CASES

Off-peak traffic produces the lowest blocking because fewer channels are occupied. Average load is intermediate. Busy-hour load is the design point: each subscriber contributes 0.025 Erlang, and the network is dimensioned for no more than 2% blocking. An event surge is the worst spatial case because demand is concentrated into a small area, not spread uniformly across the township.

For 10,000 subscribers, total offered traffic is $10{,}000 \times 0.025 = 250$ Erlangs. Under 19 evenly loaded cells, mean load is 13.16 Erlangs per cell. This is below the 38.39-Erlang limit for 48 channels at 2% blocking. At 20,000 subscribers, mean load is 26.32 Erlangs per cell, still below the same ideal limit but with less margin. At 100,000 subscribers, mean load becomes 131.58 Erlangs per cell, exceeding the per-cell limit by a factor of 3.43; blocking would rise sharply unless the network is densified or traffic is offloaded.

The aggregate capacity calculation can hide a local overload. If 30% of subscribers are concentrated in 20% of the cells, those cells carry a load density $0.30/0.20 = 1.5$ times the township average. For the 20,000-subscriber case, a hot cell would average about 39.47 Erlangs, already above the 38.39-Erlang target. This demonstrates why a population/traffic heat map is essential rather than relying only on total Erlangs.

Worst blocking occurs when a hot cell reaches or exceeds the Erlang-B design load during busy hour or event time. Dynamic borrowing can help if neighbouring cells have idle channels and the borrowed resource remains interference-safe. If all neighbours are busy, borrowing cannot create capacity. Microcells or temporary event sites are needed when the spatial demand concentration exceeds the macro layer's local channel capacity.

TASK 5 – SCALING AND SENSITIVITY

For fixed spectrum S, channels per cell scale approximately as $c \propto S/N$, so per-cell Erlang capacity decreases as N increases. Reuse distance ratio $Q=\sqrt{(3N)}$. Under the six-interferer approximation, $SIR \propto N^{(n/2)}$. The number of cells scales as $M \propto 1/R^2$. These proportionalities show the basic trade-off: larger N protects SIR but reduces per-cell channels; smaller R adds spatial capacity but increases sites, handoffs and cost.

Subscriber traffic grows linearly with population: $A=0.025U$. The four planning points are 25, 250, 500 and 2,500 Erlangs. With 19 cells, the 10,000- and 20,000-subscriber cases are supportable in the uniform ideal model at 2% blocking; 100,000 is not. At 100,000 subscribers, a first-order uniform capacity requirement is $2,500/38.39=65.1$ cell-equivalents. Thus at least 66 equally loaded N=7 cells are required by the Erlang capacity bound, before allowance for uneven traffic and coverage. Compared with 19 initial cells, this implies substantial densification.

Path-loss exponent sensitivity is not monotonic in all real environments. In the ideal formula, larger n improves co-channel isolation because distant interferers attenuate faster. At N=7, the ideal edge SIR is about 12.2 dB for n=2.5, 14.7 dB for n=3 and 18.7 dB for n=4. This explains why the supplied n=4 sanity case meets 18 dB while the open-campus n=2.5 case does not. However, dense environments may have higher shadowing, indoor penetration loss and irregular propagation, so measured performance can be worse than the simple exponent model.

A practical break-even point is around 20,000 subscribers for the selected 19-cell macro layout: it remains feasible in the uniform Erlang estimate but has little tolerance for hotspots. Beyond this point, sectorisation and targeted microcells become preferable. By 100,000 subscribers, the macro-only layout is clearly impractical. DCA helps manage uneven short-term load but cannot substitute for the additional spatial capacity needed at this scale.

TASK 5 – STRATEGY BREAK-EVEN AND SENSITIVITY

Fixed reuse is preferred at low-to-moderate demand because it has predictable interference and low operating complexity. Sectorisation becomes worthwhile when a macro cell has adequate coverage but poor SIR or high directional traffic. Dynamic assignment becomes valuable when adjacent cells have different busy-hour profiles and enough spare channels exist for safe borrowing. Cell splitting becomes necessary when the total local offered traffic exceeds the Erlang capacity of the available channels even after scheduling and sectoring.

A simple break-even test compares the cost of densification with the cost of overload. Let ΔC_{site} be the annualised cost of adding microcells and $C_{blocked}$ be the business cost of blocked calls, poor service and lost usage. Densification is justified when the avoided overload cost exceeds ΔC_{site}. The supplied brief does not give financial rates or revenue per call; therefore a numeric rupee break-even cannot be claimed. The engineering break-even can still be stated: if the forecast local load is above the 2%-blocking Erlang-B limit, a capacity change is required.

Sensitivity checks should vary R, N, n, user density, antenna pattern, shadowing standard deviation and traffic concentration. For each combination, record cell count, channels per cell, predicted SIR, Erlang-B blocking and cost. Use the same random seed in comparative simulation so that differences are due to strategy rather than unrelated random samples. A robust design should meet the target across plausible parameter ranges, not only at the nominal point.

The preferred evolution is staged. At 1,000–10,000 users, deploy macro coverage with fixed reuse and basic monitoring. Near 20,000, add sectoring at high-load sites and activate constrained DCA. As demand approaches 100,000, introduce microcells in academic, hostel and stadium hotspots, reduce local cell radius and re-optimise the reuse plan. This avoids paying for dense infrastructure in low-demand areas before it is needed.

TASK 6 – CELL-SPLITTING RECURRENCE

The initial layout has $M_0=19$ macro cells, each with radius $R_0=0.50$ km. A split replaces one parent cell with four smaller cells of radius R/2. The four-child model is an ideal area-preserving approximation because each child's area is one quarter of the parent area. The new total cell count is three more than before for each split parent: one parent is removed and four children are inserted.

For uniform splitting, every cell splits at each round. Therefore $M_k=4M_{k-1}$, with $M_0=19$, and the closed form is $M_k=19\cdot4^k$. Radius is $R_k=R_0/2^k$. Since each cell's area is one quarter per round, the number of cells per fixed township area grows by four. The ideal site counts are 19, 76, 304 and 1,216 for k=0,1,2,3 respectively. Real deployment may use fewer targeted cells because only hotspots need splitting.

Let a fraction f_k of the M_{k-1} cells be split at round k. Each selected cell adds three net cells. The selective recurrence is $M_k=M_{k-1}+3f_kM_{k-1}=M_{k-1}(1+3f_k)$. If f is constant, $M_k=M_0(1+3f)^k$. If the selected fraction varies by round, $M_k=M_0\prod(1+3f_i)$, i=1…k. This recurrence better reflects a university township where only stadium, library and hostel zones may need microcells.

Work/cost per round is proportional to the number of split parent cells. If one split costs Csplit for survey, installation and optimisation, then $W_k=Csplit\cdot f_kM_{k-1}$. Total cumulative splitting cost through round k is $\Sigma_{i=1}^{k}$ $Csplit\cdot f_iM_{i-1}$. The base case is the original 19-cell layout, not an empty network.

TASK 6 – CAPACITY, POWER AND HANDOFF RECURRENCES

If each smaller cell reuses the same local channel plan under a validated interference design, four children can collectively provide approximately four times the parent-area capacity. Let C_0 denote the initial capacity per cell-area unit. For uniform splitting, aggregate capacity over the same geographical area follows $C_k=4C_{k-1}$, so $C_k=C_0\cdot 4^k$. This is a spatial-reuse model; it assumes sufficient backhaul, radio resources and interference coordination. If the same channels cannot be reused safely at the smaller spacing, the capacity gain is lower.

The assignment gives the transmit-power relation $P'=P/2^n$. Applying it per split round gives $P_k=P_{k-1}/2^n$ and therefore $P_k=P_0/2^{nk}$. This expresses the lower power needed for a shorter propagation distance under the stated model. For $n=3.5$, one split gives $P_1/P_0=1/2^{3\cdot 5}=0.0884$; two rounds give 0.00781. This is a relative model result, not a recommended equipment power setting. Actual power is set by link budget, coverage, receiver sensitivity and interference constraints.

Handoff load rises as cells shrink. A simple first-order model assumes boundary-crossing rate is inversely proportional to cell radius for a fixed user speed and path length: $H_k\approx H_0\cdot 2^k$. Thus each split round roughly doubles handoff frequency. In practice, dwell time, road geometry, pedestrian speed and handoff thresholds matter. Excessive splitting can increase signalling and dropped-handoff risk if mobility management is not optimised.

For selective splitting, capacity is $C_k=C_{k-1}(1+3f_k)$ under the ideal four-child capacity assumption applied only to the selected fraction, and average radius is not uniform. A useful approximation is $\bar{P}_k$ and $\bar{H}_k$ tracked separately for macro and micro layers rather than forcing one radius across the township. This distinction is important for the stadium solution: temporary cells can be added locally without splitting every macro cell.

TASK 7 – SOLVING THE UNIFORM RECURRENCE

The uniform recurrence is $C_k=4C_{k-1}$, with base capacity C_0. Repeated substitution gives $C_1=4C_0$; $C_2=4(4C_0)=4^2C_0$; $C_3=4^3C_0$. By induction, $C_k=C_04^k$. To support demand D, require $C_04^k\geq D$. Divide by C_0: $4^k\geq D/C_0$. Taking logarithms gives $k\geq\log_4(D/C_0)$, so the smallest integer number of rounds is $k_{min}=\text{ceil}(\log_4(D/C_0))$.

For an illustrative capacity baseline, use $C_0=19\times38.39=729.4$ Erlangs from the 19-cell N=7 layout. At 100,000 subscribers, D=2,500 Erlangs. $D/C_0=3.427$, and $\log_4(3.427)=0.887$; therefore the ideal uniform model requires one splitting round. One round multiplies capacity by four to about 2,917.6 Erlangs, enough for the 2,500-Erlang demand in the ideal uniform model. This is only a lower-bound result: it does not address the 18 dB SIR issue in open areas, traffic hotspots, site constraints or operational margin.

The site count after one uniform round is 76. The mean offered load becomes 2,500/76=32.89 Erlangs per cell, below the 38.39-Erlang 2%-blocking limit. The capacity margin is modest: $76\times38.39=2,917.6$ Erlangs, about 16.7% above demand. A practical plan should add margin for nonuniform traffic and outages rather than deploying exactly to the mathematical threshold.

The substitution method is suitable because the recurrence is first-order and has a constant multiplier. It makes the geometric growth transparent and provides a closed form. It also shows why uniform splitting is expensive: site count grows exponentially with round number. A selective strategy should be preferred when only a small fraction of cells are overloaded.

TASK 7 – SELECTIVE SOLUTION AND MOBILITY INTERPRETATION

For constant selective split fraction f, $M_k=M_0(1+3f)^k$. If capacity scales in the same proportion, $C_k=C_0(1+3f)^k$. The minimum rounds are $k_{min}=\text{ceil}[\log(D/C_0)/\log(1+3f)]$, provided $f>0$ and $D>C_0$. If f changes by round, use the product condition $C_0\prod(1+3f_i)\geq D$ and select the smallest k satisfying it.

For example, if only 25% of cells are split in each round, f=0.25 and the per-round capacity multiplier is $1+3(0.25)=1.75$. With $D/C_0=3.427$, $k_{min}=\text{ceil}(\log(3.427)/\log(1.75))=3$ rounds. The ideal site count after three rounds is $19\times1.75^3\approx101.8$, so approximately 102 cells. This is fewer than the 1,216 sites produced by three uniform rounds, because the selective model densifies only a subset. The exact placement of those sites must follow the traffic heat map.

Power under repeated splitting follows $P_k=P_0/2^{nk}$ for uniformly split cells. For selective deployment, only split-cell transmitters follow this reduction; macro cells retain their existing power plan. Handoff frequency approximately doubles per radius-halving round in the simplified mobility model, but microcells may use handoff parameters and coverage overlap designed to reduce ping-pong. The handoff penalty is concentrated in the hotspot rather than imposed across the entire township.

The mathematical recurrence does not by itself guarantee service quality. A split requires a new site or radio node, backhaul, synchronisation, neighbour configuration, indoor/outdoor coverage design and interference validation. Capacity scaling assumes that each new cell can reuse spectrum without violating SIR. Therefore the recurrence is a planning bound; the deployment decision must be checked against measured radio conditions and lifecycle cost.

TASK 8 – SIMULATOR DESIGN AND REPRODUCIBILITY

The simulator should implement four core functions: (1) generate a hexagonal grid and cell-centre coordinates; (2) assign channels according to fixed reuse, sectoring or dynamic borrowing; (3) compute received desired/interfering power and SIR for random mobile positions; and (4) calculate Erlang-B blocking for offered traffic. Use a fixed random seed for each subscriber case so that strategy comparisons use identical user locations and shadowing samples.

Test matrix required by the brief: $N \in \{3,4,7,12\}$; $n \in \{2.5,3,4\}$; sectorised and unsectorised cases; $U \in \{1{,}000,10{,}000,20{,}000,100{,}000\}$. Report the number of cells, channels per cell, mean SIR, fifth-percentile SIR, outage probability below 18 dB, Erlang-B blocking and runtime. For a fair comparison, use the same R, grid extent, user sample and random seed for each strategy, changing only the strategy-specific parameters.

A Monte-Carlo SIR calculation samples a serving cell and user position, calculates distance to the serving site and each co-channel site, applies path loss d^{-n} and optional lognormal shadowing, then forms SIR as desired power divided by summed interference. Sectoring multiplies each received power by the antenna gain in the direction of the mobile. DCA selects only channels whose current interference estimate passes the threshold.

Validation has two layers. First, check analytical limits: the N=7, n=4 six-interferer approximation must produce approximately 18.7 dB at the cell edge. Second, test monotonicity: increasing N should improve ideal SIR; increasing N should reduce channels per cell; decreasing R should increase site count; increasing offered traffic should not reduce Erlang-B blocking. Any violation indicates a coding or modelling error.

TASK 8 – REFERENCE PYTHON IMPLEMENTATION

The following compact reference code implements Erlang-B, its inverse, the ideal hexagonal geometry, first-tier SIR and the capacity calculation. It is deliberately transparent and uses only the Python standard library. It does not claim to replace a full GIS/antenna simulator; sectoring and DCA require the additional directional and occupancy logic described in the preceding page.

```python
import math
def erlang_b(channels, traffic):
    b = 1.0
    for c in range(1, channels + 1):
        b = traffic * b / (c + traffic * b)
    return b

def inverse_erlang_b(channels, target=0.02):
    lo, hi = 0.0, float(channels * 2)
    for _ in range(80):
        mid = (lo + hi) / 2
        if erlang_b(channels, mid) > target:
            hi = mid
        else:
            lo = mid
    return (lo + hi) / 2

def valid_cluster(i, j):
    return i*i + i*j + j*j

def hex_area(radius):
    return 3 * math.sqrt(3) * radius**2 / 2

def reuse_ratio(N):
    return math.sqrt(3 * N)

def edge_sir_db(N, n, interferers=6):
    sir = (3*N)**(n/2) / interferers
    return 10 * math.log10(sir)
```

```
def plan(N, radius=0.5, area=12, spectrum=336):
    channels = spectrum // N
    cells = math.ceil(area / hex_area(radius))
    traffic_cell = inverse_erlang_b(channels, 0.02)
    return cells, channels, traffic_cell, cells*traffic_cell

for N in (3, 4, 7, 12):
    print(N, plan(N), edge_sir_db(N, 4))
```

For a full experiment, wrap the calculations in loops over U, N and n. Set `random.seed(191126)` once at the beginning, create the user sample once, and reuse it for every strategy. Store each run in a CSV file with columns `strategy, subscribers, N, n, sectors, cells, channels, mean_sir_db, p05_sir_db, outage, blocking, runtime_s`. Use a separate seed column so that the experiment can be reproduced. The assignment should include the actual console output after running this code; do not label expected values as measured results.

TASK 8 – TEST CASES AND CORRECTNESS CHECKS

Test 1 – geometry: with R=0.5 km, the computed hexagon area should be approximately 0.6495 km² and the ideal township cell count should be 19. Test 2 – reuse: N=7 should give Q≈4.5826 and D≈2.2913 km. Test 3 – cluster generation: (i,j)=(2,1) should return N=7; (2,2) should return N=12. Test 4 – Erlang-B: the inverse calculation should return approximately 38.39 Erlangs for 48 channels at 2% blocking. Substituting this traffic into Erlang-B should return 0.02 within numerical tolerance. Test 5 – SIR sanity: N=7, n=4 and six interferers should give 18.66 dB. Test 6 – monotonicity: for fixed N and n, moving the mobile closer to its serving base station must increase SIR in the distance-only model.

Test 7 – load scaling: traffic should be 25, 250, 500 and 2,500 Erlangs for 1,000, 10,000, 20,000 and 100,000 subscribers. Test 8 – channel allocation: integer channels per cell should be 112, 84, 48 and 28 for N=3,4,7,12 respectively. Test 9 – sectoring: with two equal dominant interferers instead of six, the ideal SIR should improve by 4.77 dB. Test 10 – reproducibility: identical seeds and parameters must produce identical output.

Record the Python version, operating system, package versions, seed and exact command used. Runtime comparisons are meaningful only on the same hardware and software environment. If NumPy is used for vectorisation, state its version. The reference code above intentionally avoids external dependencies, so the basic checks can run in a standard Python installation.

TASK 9 – PERFORMANCE METRICS AND TABLES

The key measurements are: (1) mean SIR in dB; (2) fifth-percentile SIR, which exposes weak-edge users; (3) outage probability P(SIR<18 dB); (4) blocking probability from Erlang-B or call-level simulation; (5) channels per cell; (6) number of sites; and (7) execution time. Report all metrics for each strategy under identical subscriber counts and propagation conditions. A single average SIR is insufficient because it can hide a poor cell-edge tail.

Analytical baseline table for N=7, R=0.5 km: area per cell 0.6495 km²; ideal cells 19; channels per cell 48; 2%-blocking offered traffic per cell 38.39 Erlangs; aggregate offered traffic capacity 729.4 Erlangs; approximate subscriber capacity 29,176. The table is an analytical model output, not a field measurement. It assumes equal loading and independent per-cell Erlang groups.

Subscribers	Total traffic (Erlangs)	Mean traffic/cell (19 cells)	Mean vs 38.39 Erlang limit
1,000	25	1.32	Below
10,000	250	13.16	Below
20,000	500	26.32	Below, limited margin
100,000	2,500	131.58	Above by 3.43×

A second table should compare N=3,4,7,12 with channels per cell, Erlang-B capacity per cell, ideal SIR at n=2.5/3/4 and site count. A third table should compare unsectorised and sectorised SIR distributions. Use the same units and decimal precision in every table. Explain whether a value is analytical, simulated or measured, and state the assumptions next to the table.

TASK 9 – PLOTS AND INTERPRETATION

Plot 1 – SIR cumulative distribution function (CDF): x-axis is SIR in dB and y-axis is the fraction of sampled users with SIR less than or equal to x. Curves further to the right indicate better SIR. Mark the 18 dB design threshold with a vertical reference line. The fifth percentile can be read from the x-value at CDF=0.05, while outage probability is the CDF value at 18 dB.

Plot 2 – blocking versus offered load: x-axis is offered traffic per cell in Erlangs; y-axis is Erlang-B blocking probability. Draw one curve for each channel count. Mark the 2% target. This graph makes the effect of N visible: larger N gives fewer channels per cell and reaches 2% blocking at a lower offered load. Plot 3 – capacity versus N: show both per-cell Erlang capacity and aggregate capacity for the fixed 336-channel pool. Label the difference clearly so the reader does not mistake a per-cell improvement for a system-wide gain.

Plot 4 – site count versus cell radius: use M=ceil(12/Ahex). The curve rises rapidly as R falls because cell area is proportional to R^2. Plot 5 – subscriber count versus blocking: compare 1k, 10k, 20k and 100k. A uniform-load curve may appear acceptable longer than the hotspot case; show both to expose traffic concentration. Plot 6 – cost versus carried Erlangs: use local cost quotations if available. If not, present a normalised cost index and explicitly label it as an assumption.

For each plot, include a descriptive title, axis units, legend, sample count and parameter values. Avoid decorative 3-D effects. A clear figure caption should explain the engineering conclusion in one or two sentences. The report's recommendation should refer back to these plots rather than relying on unsupported adjectives such as “best” or “optimal”.

TASK 10 – THEORY VERSUS SIMULATION

The analytical SIR model assumes six equidistant first-tier interferers and a mobile at the cell edge. A Monte-Carlo model samples actual user positions and actual cell-centre distances, so some users will see fewer strong interferers while others will be closer to one interferer. The mean may be higher or lower than the analytical edge estimate, but the fifth percentile is the more useful conservative comparison. The N=7, n=4 check should reproduce 18.66 dB only when the simulation is configured to match the same six-equidistant geometry.

The analytical Erlang-B model assumes Poisson call arrivals, independent exponential holding times, a fixed number of circuits and no waiting queue. A simulation may include finite user populations, non-Poisson bursts, variable call durations, handoff reservations, retries and priority traffic. These effects change blocking. A finite-tier grid also has fewer interferers at the network boundary than an infinite repeating lattice, which can make simulated SIR appear artificially high near the township edge.

Shadowing causes random variation in both desired and interfering paths. A lognormal shadowing model may widen the SIR distribution and increase outage even if the mean remains similar. Integer rounding of N and channels per cell creates discontinuities. Sector antenna patterns and DCA occupancy also violate the simple equal-power six-interferer assumption. These are legitimate reasons for differences, not necessarily coding errors.

The comparison table should contain analytical SIR, simulated mean SIR, simulated fifth percentile, analytical Erlang-B blocking, simulated blocking and absolute difference. Explain whether the model is optimistic or conservative for each metric. If simulation does not match the known sanity case, first check distance units, dB conversion, exponent sign, interference summation and whether the serving signal was accidentally included as interference.

TASK 10 – STRATEGY SCALABILITY CONCLUSION

Fixed reuse is the strongest baseline for predictable interference and low complexity. Sectorisation is the most direct way to improve SIR in a directional hotspot without increasing spectrum. DCA is strongest when demand is uneven and neighbouring cells have idle interference-safe resources. Hierarchical microcells are essential when the total demand exceeds what the macro layer can carry, because they create additional spatial reuse. The scalable solution is therefore a hybrid, not a single isolated strategy.

For the 19-cell N=7 base layout, the ideal aggregate 2%-blocking capacity is about 729 Erlangs, corresponding to approximately 29,176 subscribers at 0.025 Erlang each. At 100,000 subscribers, demand is 2,500 Erlangs. The analytical uniform split bound gives one round of four-way splitting, 76 cells and about 2,918 Erlangs of ideal capacity. However, a targeted design can reduce cost by splitting only cells where the traffic map shows overload.

Theory and simulation should be used together. Theory explains why N, R, n and interferer count matter; simulation reveals spatial variation, shadowing and nonuniform occupancy. Neither alone is sufficient. A simulation should not be used to hide an analytically impossible capacity plan, and an analytical approximation should not be treated as a substitute for measured propagation. The final recommendation must satisfy SIR, blocking, coverage, mobility and cost constraints simultaneously.

The preferred design for the township is a macro fixed-reuse layer with N=7 as the initial conservative plan, targeted three-sector sites where measured SIR or directional demand requires them, and constrained DCA for local load balancing. As demand grows, add microcells selectively in academic/hostel hotspots. This preserves a stable baseline while allowing capacity to follow real demand.

TASK 11 – FINAL NETWORK DESIGN

Recommended initial design: 19 ideal macro cells over 12 km², nominal R=0.50 km, N=7, 48 net traffic channels per cell, and n=3.5 as the planning exponent with sensitivity tests at n=2.5 and n=4. Use 18 dB as the target SIR. The ideal N=7, n=4, six-interferer edge result is 18.66 dB; for n=3.5 it is only about 16.9 dB. Therefore the unsectorised N=7 design cannot be claimed to meet 18 dB everywhere across open-campus conditions. Sectorise or adjust reuse after propagation survey.

Capacity: 48 channels support approximately 38.39 Erlangs per cell at 2% blocking. Nineteen evenly loaded cells support about 729.4 Erlangs, or roughly 29,176 subscribers at the stated traffic rate. This supports the 1,000, 10,000 and 20,000 cases under uniform loading; it does not support 100,000. The 100,000 case requires at least 66 N=7 cell-equivalents by the Erlang capacity bound, and more may be needed for hotspots, edge coverage and operational reserve.

The recommended evolution is staged: (1) deploy the macro layer and conduct drive tests; (2) apply 120° sectoring at high-traffic or poor-SIR sites; (3) enable interference-constrained DCA/borrowing; (4) add microcells in the busiest academic, hostel and arterial-road locations; (5) use a dedicated temporary stadium layer during major events. Keep a macro coverage umbrella to support mobility and rural/open-space continuity.

This design is superior to a pure fixed-reuse network because it adapts to directional and time-varying demand. It is superior to pure DCA because it retains a predictable interference-safe baseline. It is more economical than uniform cell splitting because new sites are concentrated where demand requires them. Its limitations are higher operational complexity, additional handoffs and the need for accurate radio measurements.

TASK 11 – COST, TRADE-OFFS AND LIMITATIONS

The cost plan must include initial traffic estimation, propagation survey, candidate-site search, drive testing, hex-grid planning, site acquisition, radio equipment, antennas, power, backhaul, controller/software, installation and optimisation. Annual cost includes rent/lease, electricity, backhaul, maintenance, software support and periodic drive tests. Since no local cost data are provided in the brief, the correct report uses a cost model and obtains quotations rather than fabricating rupee amounts.

The principal trade-off is trunking efficiency versus interference protection. Small N gives more channels per cell but shorter reuse distance and potentially poorer SIR. Large N improves separation but leaves fewer channels per cell. Sectoring reduces dominant interferers but adds antennas, radio chains and sector handoffs. DCA improves utilisation but increases signalling and coordination. Cell splitting adds capacity but increases site count, power/backhaul requirements and handoff load.

The design remains practical beyond 100,000 subscribers only as a layered, densified network. The original 19-cell macro plan is not sufficient at that scale. The ideal uniform recurrence suggests 76 cells after one four-way split round, but a selective design may be cheaper if it targets hotspots. The final number must be established by a spatial traffic map and measured propagation. If the subscriber forecast exceeds the ideal capacity margin, another split round or additional microcells are required.

Limitations: ideal hexagons do not match real terrain; the path-loss exponent is a simplification; the first-tier SIR approximation ignores distant interferers and antenna sidelobes; Erlang-B assumes stationary random arrivals; subscriber traffic may be correlated; and the assignment provides no site cost, call duration, antenna pattern or measured survey data. These limits are explicitly stated so that the recommendation is technically honest and can be validated in a field trial.

TASK 12A – EVENT IMPACT ANALYSIS

The event brings 25,000 attendees at 0.05 Erlang each. Total event traffic is 25,000×0.05=1,250 Erlangs concentrated in 1 km² for four hours. Event traffic density is 1,250 Erlangs/km². The normal 20,000-subscriber township case carries 500 Erlangs across 12 km², or 41.67 Erlangs/km² on average. The event zone therefore has about 30 times the average traffic density of that 20,000-subscriber baseline. Compared with the 100,000-subscriber average township density of 208.33 Erlangs/km², the event density is about six times higher.

The event attendees' average per-person traffic is double the normal subscriber assumption (0.05 versus 0.025 Erlang). The event zone also violates the uniform-traffic assumption: a large share of users occupy a small area. It violates steady-demand and permanent-capacity assumptions because the surge lasts four hours and occurs roughly twice per month. The short, predictable duration matters: a temporary capacity layer may be more economical than permanently densifying the entire stadium zone for only eight event-hours per month.

The stadium zone covers 1 km². With R=0.5 km, its ideal hexagonal area is 0.6495 km², so at least ceil(1/0.6495)=2 cell-equivalents are needed geometrically. But two cells each with 48 channels can carry only about 76.78 Erlangs at the 2% target, far below 1,250 Erlangs. Even if all 19 macro cells' aggregate capacity were available in the stadium zone, it would be an unrealistic assumption because the rest of the township still has ordinary traffic.

The changed requirement therefore makes the original unchanged macro plan unacceptable for the stadium. The first failure is local channel exhaustion and blocking, followed by increased co-channel interference if channels are borrowed without coordination. The event must be handled by temporary capacity concentration, traffic steering and a measured interference plan.

TASK 12B – EVALUATION OF ORIGINAL DESIGN

The normal 19-cell N=7 plan provides 48 channels per cell and about 38.39 Erlangs per cell at 2% blocking. The stadium zone has at least two macro cell-equivalents, giving a simplified local capacity of 2×38.39=76.78 Erlangs. Event load is 1,250 Erlangs, which is 16.28 times this local 2%-blocking capacity. This comparison is conservative in that it assumes both cells are fully dedicated to the stadium; in reality they also serve nearby residents and campus buildings.

The local offered load per two cells would be 625 Erlangs per cell if evenly split, compared with the 38.39-Erlang 2%-blocking design point. The exact blocking probability at this overload is close to saturation and should be calculated with the Erlang-B recurrence for the relevant channel count; the 2% target is clearly violated. A precise call-level blocking value should not be over-interpreted because the 25,000 attendees may not all initiate calls independently and traffic may be bursty.

Interference risk increases if the network responds by borrowing channels from distant cells or increasing macro transmit power. Borrowing is safe only when the donor and all co-channel neighbours maintain the SIR threshold. Increasing power indiscriminately raises interference to surrounding cells and does not create additional channels. Sectorisation alone is also insufficient: it improves SIR but does not multiply the licensed traffic channel pool enough to carry 1,250 Erlangs in the stadium zone.

A temporary cell-on-wheels (COW), portable microcells or deployable small-cell cluster is more economical than permanent network-wide redesign. These temporary nodes should use coordinated frequency/resource assignments, directional antennas, adequate backhaul and a macro overlay. Dynamic borrowing can complement the temporary layer for small residual imbalances but should not be the primary capacity source.

TASK 12C – ADAPTATION / REDESIGN

Decision: MODIFY the original design using a temporary stadium capacity layer, retaining the macro network for ordinary township coverage. Deploy temporary microcells or a COW cluster around the stadium before the event, with sectors directed into the crowd area. Use a dedicated local reuse plan and controlled power. Keep the 19-cell macro layer active for mobility, emergency coverage and nearby non-event users. Activate constrained DCA only after the temporary layer is operating and interference checks pass.

A first-order capacity requirement is 1,250 Erlangs at 2% blocking. With 48 traffic channels per temporary cell, each cell supports about 38.39 Erlangs, so the ideal number of fully dedicated cell-equivalents is ceil(1,250/38.39)=33. This is a theoretical lower bound under independent Erlang groups and equal load. A real event design needs additional margin for uneven crowd distribution, handoffs, uplink limitations, data demand and safety services. If each temporary sector has fewer than 48 effective channels, the required number rises.

The revised layer should use 120° or 60° sectoring depending on site layout and measured antenna patterns. Sectorisation reduces dominant interferers and helps direct energy into the stadium. Microcell radius is reduced locally; lower transmit power limits spillover and improves reuse. Handoff boundaries are concentrated around the event perimeter. Temporary nodes should be installed and tested before gates open, then removed or powered down after the event.

Capacity effect: the temporary layer adds spatially reusable local capacity without increasing the licensed spectrum. SIR effect: directional antennas, reduced cell radius and coordinated channel assignment reduce dominant interference, but the actual result must be verified by simulation and drive tests. Cost effect: event-specific rental, transport, installation, backhaul and engineering are incurred about twice per month, avoiding permanent sites that would be underused most of the time.

TASK 12D – EVIDENCE-BASED DEFENCE

The modify decision follows from the numerical mismatch. Event demand is 1,250 Erlangs in 1 km², while two baseline macro cell-equivalents provide only about 76.78 Erlangs at the 2% blocking point. The demand-to-capacity ratio is 16.28. Retaining the original design unchanged would violate the blocking target. Permanent redesign of the whole township would impose recurring cost for a surge that occurs only about twice monthly and lasts four hours per event.

A temporary layer is the balanced response. The ideal lower-bound estimate of 33 cells with 48 channels each shows the scale of the event capacity problem. The final deployment count should be determined by the actual temporary cell channel plan and a simulation with crowd distribution. Sectoring improves SIR but does not alone solve the capacity deficit; DCA helps only where spare interference-safe channels exist. Therefore the temporary microcell/COW layer is the primary intervention, with sectoring and DCA as supporting mechanisms.

The cost comparison should use event rental and deployment cost for approximately 8 event-hours per month against the annualised cost of permanent stadium microcells. The assignment supplies no monetary rates, so the report does not invent a rupee break-even. The qualitative conclusion is strong: because the event is predictable and intermittent, temporary capacity is likely to have better utilisation and lower lifecycle cost than permanent full-scale densification, provided setup and testing are reliable.

Operational controls include pre-event radio survey, temporary-site permissions, backhaul capacity, power backup, emergency-service priority, crowd-area coverage testing, interference monitoring during the event, and post-event removal. A fallback plan should retain macro coverage if a temporary node fails. This defence combines traffic arithmetic, Erlang-B capacity, SIR reasoning, event duration and lifecycle-cost logic.

OVERALL CONCLUSION AND DEPLOYMENT ROADMAP

The recommended township solution is a layered hybrid. Begin with a 19-cell ideal macro layout at R=0.50 km and N=7, with 48 net traffic channels per cell. Validate the 18 dB SIR target through propagation surveys; where the n=2.5–3 open-campus conditions do not meet the ideal target, use sectoring, antenna optimisation or a revised reuse plan. Use DCA only with interference-aware admission and borrowing rules. Add microcells selectively as demand approaches and exceeds the macro-layer capacity.

The 19-cell baseline has an ideal aggregate 2%-blocking capacity of approximately 729 Erlangs, or 29,176 subscribers at 0.025 Erlang each. It is sufficient for 1,000 and 10,000 users and nominally supports 20,000 users under uniform load, but has limited hotspot margin. It is not sufficient for 100,000 users. A uniform four-way split gives 76 cells and an ideal capacity near 2,918 Erlangs, while selective splitting can concentrate new sites in overloaded zones at lower cost.

The event challenge requires a separate response: 1,250 Erlangs in 1 km^2 is far above the local macro capacity. Modify the design temporarily with a stadium microcell/COW cluster, sectorised antennas and controlled DCA. Keep the macro layer for coverage and mobility. This avoids permanently overbuilding the whole township for a short, predictable event.

The central engineering lesson is that frequency reuse, interference, traffic and site density must be planned together. No single parameter can be maximised independently. A sound design balances SIR, Erlang capacity, mobility, capital cost, operating cost and growth. Final deployment numbers must be confirmed with real propagation measurements, detailed traffic maps, antenna patterns and local cost quotations.

REFERENCES, FORMULAE AND SUBMISSION CHECKLIST

Core formulae used: hexagonal area $A_{hex}=(3\sqrt{3}/2)R^2$; valid cluster size $N=i^2+ij+j^2$; reuse distance $D=R\sqrt{(3N)}$; reuse ratio $Q=\sqrt{(3N)}$; channels per cell $c=\text{floor}(S/N)$; Erlang-B $B(c,A)=A\cdot B(c-1,A)/(c+A\cdot B(c-1,A))$; first-tier edge SIR$\approx$(3N)^(n/2)/I, where I is the effective number of equal dominant interferers; and cell count $M=\text{ceil}(A_{region}/A_{hex})$.

Recurrences: uniform splitting $M_k=4M_{k-1}$, $C_k=4C_{k-1}$, $R_k=R_0/2^k$, $P_k=P_0/2^{nk}$, and approximate handoff rate $H_k\approx 2^k H_0$. Selective splitting $M_k=M_{k-1}(1+3f_k)$, with corresponding ideal capacity scaling only for the fraction split. Uniform minimum rounds are $k=\text{ceil}(\log_4(D/C_0))$; constant selective fraction gives $k=\text{ceil}[\log(D/C_0)/\log(1+3f)]$.

Reference concepts: cellular frequency reuse and hexagonal geometry; Erlang-B loss-system traffic modelling; path-loss and lognormal shadowing; sector antenna patterns and spatial filtering; dynamic channel assignment and trunking; hierarchical macro/microcell planning; and Monte-Carlo simulation methodology. These are standard communications-engineering concepts. The numerical inputs and assignment-specific constraints are taken from the supplied C01 assignment brief.

Before submission, run the reference code and insert actual output/screenshots for Task 8; generate the requested CDF and blocking plots for Task 9; record machine/software details; and replace the cost-model variables with local quotations if available. Confirm that the final Word document's page count remains within the required 40–45 pages after any edits. Add the actual submission date and verify the student details on the cover page.